

TrES-2b

R-1662

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_130926 · created 2026-08-02T09:34:46+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456561.66241 +/- 0.00025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.148 +/- 0.034
Transit depth (Rp/Rs)^2	2.19 +/- 1.0 [%]
Orbital Inclination (inc)	82.75 +/- 1.13
Airmass coefficient 1 (a1)	1.0368 +/- 0.0099
Airmass coefficient 2 (a2)	-0.033 +/- 0.012
Scatter in the residuals of the lightcurve fit is	0.95 %
Best Comparison Star	None
Optimal Aperture	3.97
Optimal Annulus	7.65
Transit Duration (day)	0.05 +/- 0.03

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.148 ± 0.034 vs 0.1254 (deviation 0.42σ)
Duration fit vs published	0.05 d vs 0.0746 d (deviation 0.52σ)
Mid-time O–C	0.6 min
Depth SNR	2.7
Residual scatter	0.95 %

Light curve

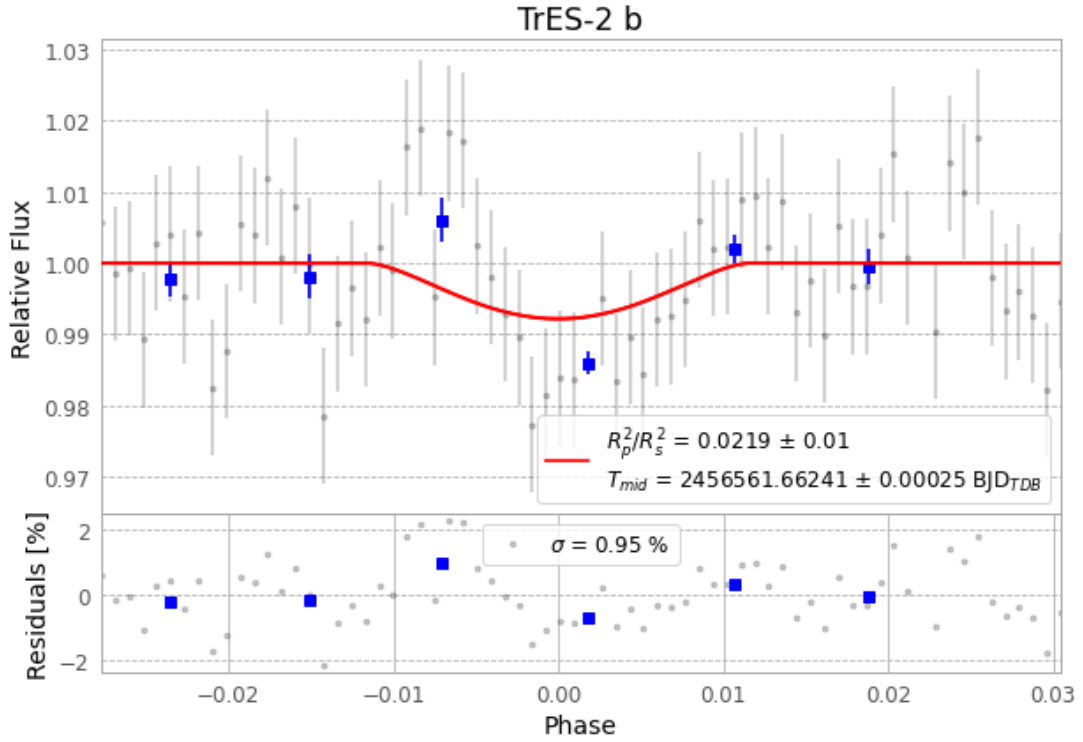


Figure 1 — FinalLightCurve_TrES-2 b_2013-09-25.png (SHA-256 865ec05b5fb58145...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [408, 251], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 f0a21941493b1b4f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

70 FITS frames (46 MB), TRES-2130926021212.FITS → TRES-2130926053912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-130926.log (5f8a0ffbc8ee...), FinalLightCurve_TrES-2 b_2013-09-25.png (865ec05b5fb5...), FinalLightCurve_TrES-2 b_2013-09-25.csv (06136e892bbe...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 09:34Z.